

1 Introduction

Viterbo’s conjecture asserts that among convex bodies of fixed volume the ball has the largest symplectic capacity [Vit00]. We work in $\mathbb{R}^4$ with the Ekeland–Hofer–Zehnder capacity c_{EHZ} : for a convex body K it is the least action of a closed characteristic on ∂K , where for polytopes the closed characteristics are replaced by generalized characteristics. The *systolic ratio*

$$\text{sys}(K) = \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}$$

equals one for the ball, and the conjecture asserts $\text{sys}(K) \leq 1$ for every convex body $K \subset \mathbb{R}^4$. Haim-Kislev and Ostrover disproved it with a Lagrangian product [HO26]. For planar convex bodies K_q and K_p , the Lagrangian product $K_q \times_L K_p$ places K_q in the (q_1, q_2) -plane and K_p in the (p_1, p_2) -plane. With P a regular pentagon and $R(\theta)$ the rotation by θ , the HKO body $K_{\text{HKO}} = P \times_L R(-\pi/2)P$ has systolic ratio $(3 + \sqrt{5})/5 \approx 1.047$.

This thesis fixes a toolbox and tests how far it gets on questions around Viterbo’s conjecture after this counterexample. The toolbox consists of high-performance computation on Augsburg’s LICCA cluster, proof by computation in exact arithmetic, standard data-science methods, and AI agents. The agents ran the data-science experiments and did large parts of the theory work; Section ?? describes how. The questions were not chosen first: they came out of the work.

Standard data science found no new body with systolic ratio above one. In random samples of generic polytopes and of Lagrangian products of polygons, and in local searches started from generic ten-facet polytopes, every computed systolic ratio is below one. Perturbations of K_{HKO} have systolic ratio above one, but they are small variations of the known counterexample. The data did show a strong inverse association between systolic ratio and symplectic ridge area, the total symplectic area of the two-dimensional faces normalized by volume. In knot theory, patterns found in computed data have suggested results that then inspired theory; here AI agents both look for the patterns and do much of the theory.

Randomly sampled perturbations of K_{HKO} never increased its systolic ratio. This raised the first question that the thesis answers by a theorem: is K_{HKO} a local maximum of the systolic ratio? We prove that it is among polytopes with exactly ten facets, and that the same space contains a second local maximum, a hexagon–quadrilateral product with systolic ratio one. Other results answer questions that arose the same way: an explicit formula for the systolic ratio of $P \times_L R(\theta)P$ as a function of θ , the sharp bound $(3 + \sqrt{5})/5$ for products of independently linearly deformed regular pentagons, and a capacity-increasing rotation of any product of planar convex bodies out of symplectic position. The capacity computations rest on Haim-Kislev’s finite formula [Hai19], a reduction for Lagrangian products, and a second algorithm built on the local geometry of Chaidez and Hutchings [CH21].

The subsections below first recall how capacities connect embeddings, dynamics and convex geometry, then introduce the local theorem, the finite capacity calculations and the data-science results, and end with suggested reading orders.

References

- [CH21] Julian Chaidez and Michael Hutchings. “Computing Reeb dynamics on four-dimensional convex polytopes”. In: *Journal of Computational Dynamics* 8.4 (2021), pp. 403–445. DOI: 10.3934/jcd.2021016.
- [Hai19] Pazit Haim-Kislev. “On the symplectic size of convex polytopes”. In: *Geometric and Functional Analysis* 29.2 (2019). arXiv:1712.03494, pp. 440–463. DOI: 10.1007/s00039-019-00486-4.

- [HO26] Pazit Haim-Kislev and Yaron Ostrover. “A counterexample to Viterbo’s conjecture”. In: *Annals of Mathematics* 203.2 (2026), pp. 603–622. DOI: 10.4007/annals.2026.203.2.5.
- [Vit00] Claude Viterbo. “Metric and isoperimetric problems in symplectic geometry”. In: *Journal of the American Mathematical Society* 13.2 (2000), pp. 411–431. DOI: 10.1090/S0894-0347-00-00328-3.